

Accumulated environmental impact: the case of cadmium in Sweden

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Abstract

In this study, the total flows of cadmium in Sweden in the period 1940–1990 — based on trade statistics, the manufacture of goods and the persistence of products in the environment — have been calculated. The metal industry, the mining of zinc and lead ores and the manufacturing of phosphorus fertilizers have been the dominant sources of industrial cadmium emissions to the environment. The application of fertilizers has led to the depositing of significant amounts of cadmium on agricultural land. Consumption emissions have originated from the use of cadmium in various products, e.g. rechargeable nickel-cadmium batteries, pigments, stabilizers in polyvinyl-chloride plastics and protective plating for metals. The total calculated emissions of cadmium in Sweden, from production and consumption, have, in the past, been approximately 1700 t. The accumulated amount of cadmium used, including cadmium in alloys and as impurities in zinc, is approximately 5000 t. The ‘societal weathering rate’ exceeded the natural rate more than 4 times in 1970, and the present rate (1990) of anthropogenic emissions is still higher than the natural release due to weathering.

Key words: Cadmium; Consumption; Emission; Technosphere

1. Introduction

Man has been a consumer of heavy metals, such as lead and silver, in various applications for thousands of years. However, the use of cadmium has a short history. It was discovered in the 19th century and, in Sweden, the amounts used before the Second World War were limited. The major uses of cadmium have been rechargeable nickel-cadmium (Ni-Cd) batteries, pigments, stabilizers

in polyvinyl-chloride plastics and protective plating for metals. The metal industry, the mining of zinc and lead ores and the manufacturing of phosphorus fertilizers have been the dominant sources of industrial cadmium emissions to the environment. The application of fertilizers has led to the depositing of significant amounts of cadmium on agricultural land. However, the total flows of cadmium in society and the environment are not of the same magnitude as other heavy metals (e.g. chromium and lead). However, even if the amounts of anthropogenic cadmium are small, the

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toxicity and the mobility of cadmium will constitute an environmental problem in the future. Clearly this problem will become even more accentuated as a result of ongoing acidification.

Cadmium is a rare element. Estimates of its abundance in the earth's crust range from 0.1 to 0.2 ppm, making it the sixty-seventh element in order of abundance (Bewers et al., 1987). Because there are no separate ores of cadmium, at least none of commercial importance, cadmium is produced exclusively as a by-product, mainly in the recovery of primary zinc from its ores, from zinc-bearing lead ores, or in the processing of secondary materials, e.g. scrap metal. The main producers were the USA, the former USSR and Japan. Even if the world production of cadmium increased from 5000 t in 1940 to approximately 22 000 t in the 1980s (SOU 1990:59), the total amounts of cadmium used in society are small compared with most other heavy metals. However, as organisms are adapted through evolution to low environmental concentrations, even small anthropogenic contributions will be hazardous.

Cadmium is a silver-white metal which is ductile and easily worked; it can be rolled into sheets and drawn into wire and it is easily soldered. Cadmium appears naturally as Cd^{2+} , often in complexes with inorganic (e.g. Cl^- , F^-) or organic ligands. In soil, the solubility of CdCO_3 and possibly $\text{Cd}_3(\text{PO}_4)_2$ controls cadmium mobility (Kabata-Pendias and Kabata, 1984). However, the solubility of cadmium is highly dependent on the pH and in acidic soils cadmium is one of the most mobile of heavy metals.

Cadmium is a highly toxic element responsible for acute and chronic toxicity in man (Margeli et al., 1991). On average, 5% of ingested cadmium is absorbed (GESAMP, 1986) and about one-third of this is transported to the kidneys. Its biologic half-life is approximately 20 years, which means cadmium is accumulated in the body. In Sweden, the average cadmium uptake from food is $15\mu\text{g}/\text{day}$ (Bingman, 1987), i.e. a quarter of the WHO recommended daily intake ($60\mu\text{g}/\text{day}$). However, individual variation in uptake coupled with exposure from other sources, especially cigarettes, may give rise to a higher total.

The first sign of cadmium intoxication is the

dysfunction of the kidneys in the form of decreased renal tubular absorption of proteins. This is the critical effect of long-term exposure to cadmium. If this could be prevented, even more serious effects such as mutagenicity, carcinogenicity and reproductive damage could be avoided.

To understand the potential risks of cadmium it is necessary to obtain a knowledge of total emissions, flows and accumulation in the past and the present and to make estimates for the future. In this study, the total flows of cadmium in Sweden in the period 1940–1990, based on trade statistics, the manufacture of goods and the persistence of products in the technosphere, have been calculated. Industrial emissions (we refer to these as production emissions) have been limited to relatively few sources. However, large amounts of cadmium have been used in various products from which significant quantities have been released into the environment through what we refer to as consumption emissions.

In the following reconstruction of historical emissions, general terminology and technical procedure have been adopted from previous studies on chromium and lead (Anderberg et al., 1989,1990; Bergbäck et al., 1989,1992).

2. The use of cadmium in Sweden

As there has been no major zinc/cadmium production in Sweden, nearly all the cadmium used in the country is imported. Cadmium has five principal uses: (i) in pigments, (ii) in stabilizers, (iii) in Ni-Cd batteries, (iv) as protective plating on steel, and (v) in various alloys (Table 1). Since the early 1980s, the use of cadmium in pigments, stabilizers and plating has been restricted in Sweden by legislation.

The sulphide of cadmium is used extensively in pigments. It can produce colours from very light yellow, through orange and light red to deep maroon, either on its own or with varying amounts of cadmium selenide. About 90% of cadmium pigments are used in plastics, mainly polyvinyl chloride (PVC), polythene and polystyrene. Most of the remaining 10% are used in glass and ceramics. The use of cadmium pigments in Sweden has been estimated by the Swedish Environmental

Table 1
Estimated use of cadmium in Sweden 1940–1990, (t/year)

	Pigments	Stabilizers	Ni-Cd batteries		Plating	
			Jungner	Sintered-plate		
1940	—	—	3	—	20	
1945	—	—	5	—	20	
1950	0.3	1	12	—	25	
1955	1	2	10	—	30	
1960	3	6	12	—	35	
1965	6	13	17	—	40	
1970	13	29	16	1	40	
1975	18	40	25	6	30	
1980	7	30	21	18	20	
1985	0.5	9	20	24	2	
1990	0.5	2	11	90	2	
Total (1940–1990)	240	660	710	525	1280	3415

Protection Board (SNV) from 1977 onwards (Mörtstedt, 1979; Bothen and Fallenius, 1982; S. Olsson, SNV, pers. commun.). In calculating backwards in time, the consumption of cadmium in pigments was assumed to follow the production of PVC.

Organic cadmium salts (notably cadmium stearate and cadmium benzoate) have been used as stabilizers, mainly for clear PVC plastics. These stabilizers protect plastics from degradation caused by heat and light. The use of cadmium in stabilizers from 1977 (Mörtstedt, 1979) was used as a starting point for calculating backwards using the production of PVC as a controlling index. For the period 1978–1990 different estimates made by Bystedt (1978, 1989) and Flensjö (1983) were used.

There are two distinct categories of Ni-Cd storage batteries: the pocket type (Jungner type) and the sintered-plate type. The bulk of cadmium metal imports to Sweden are used in the production of Jungner-type batteries. However, approximately 90% of these batteries have been exported. The domestic use of cadmium for this application has been estimated at 20 t/year (SOU 1990:59) in the mid-1980s. Earlier domestic use has then been calculated according to the net cadmium-metal consumption.

The use of the sintered-plate type of Ni-Cd rechargeable cells in portable appliances and tools increased drastically in the 1980s (SOU 1990:59). Since 1985 a 30% yearly increase in consumption has been assumed to match the expansion in production (e.g. in Japan). For the period 1970–1985 sale statistics and estimations of the Swedish Environmental Protection Board have been used (Kjällman et al., 1983).

Electroplating of various metals and alloys has been one of the most important uses of metallic cadmium, where it provides a coating that is resistant to corrosion by alkalies, salt water or the atmosphere. In the early 1970s the consumption of cadmium for this use was 40 t/year (Mörtstedt, 1979), but in 1978 it was reduced significantly to 2 t/year. The use of cadmium in the period 1940–1970 has been calculated roughly in relation to the number of electroplating industries established (National Swedish Environmental Protection Board, 1972).

Cadmium has extensive use in alloys. For example, cadmium is alloyed with copper to improve its strength and wear resistance (e.g. car radiators). The consumption of cadmium in domestically produced alloys was approximately 10–40 t/year between 1940 and 1990 (Mörtstedt, 1979; Bingman,

1987). However, most of this amount has been exported in the form of various products. Thus, the total amount of cadmium used in Sweden can be estimated at 5–20 t/year, i.e. 250–1000 t for the whole period. The import of cadmium in alloyed products has not been considered due to the lack of reliable data. The Swedish consumption of zinc in the mid-1980s was approximately 50 kt/year. The total amount of zinc used in Sweden in the period 1940–1990 was 2.1 million tonnes (SOU 1979:40). As there has been no production in Sweden, all zinc has been imported. In more recent decades this has mainly been imported as electrolytic zinc with an average cadmium content of 0.001%. This amounts to a total of 20 t of cadmium for the period 1940–1990. In a study of zinc production in the USA in 1968, an average content of 0.01% was found when considering different types of zinc processed (special high, 0.002; prime western, 0.025%) (Fulkerson and Goeller, 1973). For Sweden, this would amount to a total of 200 t of cadmium for the period studied. As shown in Table 1, Ni-Cd batteries (the sintered-plate type) is the only use that is extensive. Nevertheless, this increase is quite remarkable.

3. Calculation of emissions

Estimates have been made for the most important industrial branches, the use of fertilizers and consumer uses. The cadmium emissions from various sources have been estimated for the period 1940–1990. This 50-year period reflects nearly all anthropogenic emissions in Sweden with one major exception: cadmium in phosphorus fertilizers. Thus, to include this part, the accumulated amounts of cadmium in agricultural soils have been calculated from the beginning of this century.

The main principle for production emissions has been to use the earliest available estimations for branches of industry and then let the emissions follow the development of production and/or cadmium use backwards in time. The emissions of a branch of industry have been distributed between various factories according to the number of workers employed or the production figures at different times. Finally, the total emissions per time period and administrative region (there are 70 A-

regions in Sweden, cf. Fig. 1) have also been calculated.

For consumption emissions specific factors for various products have been used, and the emissions have been distributed between A-regions according to the distribution of population, except in the case of fertilizers where the acreage of arable land has been used.

3.1 Production emissions

The national emissions have been estimated by the Swedish Environmental Protection Board (SNV) for various heavy metals and branches of industry since the beginning of the 1970s. These

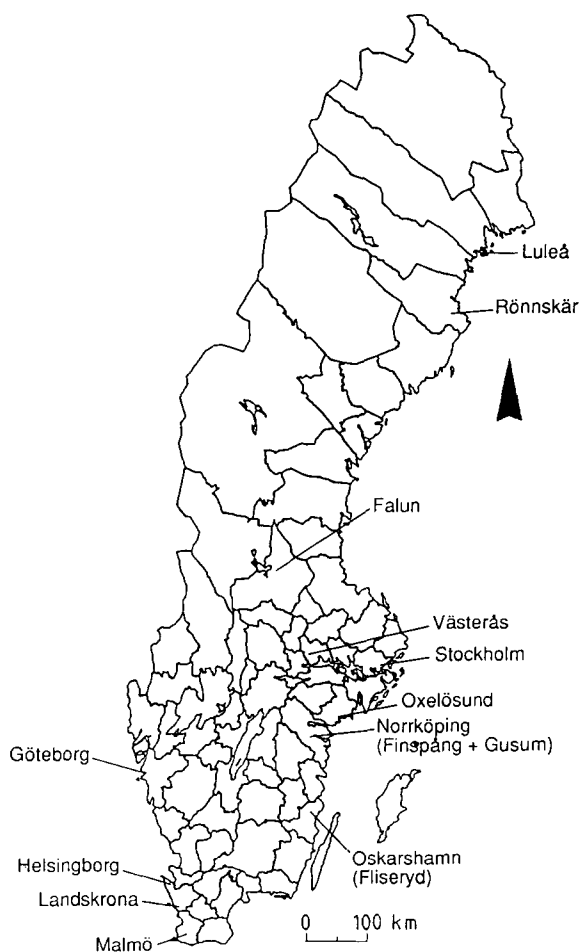


Fig. 1. Swedish administrative regions (A-regions). Cities and some point sources mentioned in the text are marked.

estimates have been used for the calculation of emissions from the most important industries backwards in time. Water emissions have been calculated for metalworks, iron and steelworks, mining and the manufacture of batteries, phosphoric acid and phosphorus fertilizers. These branches of industry accounted for 97% of emissions at the end of the 1970s (sewage treatment plants excluded). Atmospheric emissions have originated from metalworks (primary and secondary), iron and steelworks and battery manufacturing (93% of total production emissions to air in 1977) (Table 2). Note that refuse incineration and cable burning are considered in this study as consumption emissions.

3.1.1. Iron and steel and non-ferrous metals. Modern metallurgy expanded in Sweden during the interwar period. Official estimates of emissions have been used for the most recent decades and emissions have been assumed to follow production backwards in time. For the iron and steel industry the consumption of hard coal and coke was used for the calculation of air emissions and the production of raw steel for water emissions. Coke making was included for two plants, Oxelösund and Luleå, by using emission factors from ERL (1990) and the production of coke ($1\mu\text{g Cd/kg coal}$; proportional

emissions: 0.14 air, 0.19 water, 0.67 product coke). Primary non-ferrous metal production has been totally dominated by the smelter at Rönnskär in northern Sweden, which started in 1933. Before 1970 cadmium emissions have been assumed to follow silver production. Emissions from secondary non-ferrous metal production (mainly copper) have been estimated from secondary copper production. Approximately 60% of these emissions have originated from Finspång, the rest have been divided between Gusum (the Norrköping region) and Skultuna (Västerås).

3.1.2. Phosphorus fertilizers and phosphoric acid. Water emissions from the manufacture of phosphorus fertilizers and phosphoric acid have been calculated from the production of P_2O_5 . An emission factor of 58 kg Cd/t P_2O_5 was used. One-third of this cadmium has ended up in the residual product gypsum, which has been emitted directly into the water recipient (e.g. Landskrona). However, in Helsingborg some parts of the gypsum have been dried and used in construction material over the last two decades. This potential cadmium source has not been taken into consideration here.

3.1.3. The manufacture of Ni-Cd batteries. There has been no production of sintered-plate type batteries in Sweden, but pocket Ni-Cd batteries have been manufactured in Fliseryd/Oskarshamn (southeast Sweden) since 1910. The emissions have been calculated from the use of cadmium and the production of batteries. The production of cadmium mass in Fliseryd (1910–1974) has given significant total emissions to both air and water (8 and 24 t, respectively).

3.1.4 Total production emissions. The estimations of industrial emissions to water are summarized in Table 3 and Fig. 2. Total emissions for the period 1940–1990 amount to 250 t. In the 1950s the production of fertilizers was a major source of water emissions, but total emissions were dominated by iron and steel production (24%) and non-ferrous metallurgy (28%). For the mid-1980s emissions from mining areas constitute a major proportion. However, estimations of these emissions, both from mining activities and leaching from mine waste deposits, are uncertain. Here, we have used estimations from the 1970s and 1980s (Qvarfort, 1979; Ahl et al., 1983) to represent the

Table 2
Cadmium emissions to air and water in Sweden (1977) estimated by the Swedish National Environmental Protection Board (SNV), (t/year)

	Air	Water
Mining	0.2	1.4 ^a
Metal works	6	1.8
Iron and steel works	2	0.1
Cable burning ^b	0.5	—
Refuse incineration ^b	3	—
Production of phosphoric acid and fertilizers	—	0.5 ^c
Other sources	0.4	0.1

^aIncluding leakage from old mining waste.

^bCable burning and refuse incineration have here been considered as consumption emissions.

^c0.3 t from production of phosphoric acid (Granath, 1978). Liedholm et al. (1981) give a yearly emission of 0.9 t, which probably is an overestimate.

Table 3

Calculated cadmium emissions to water from various branches of industry in Sweden 1940–1990, based on production figures (t/year)

	Non-ferrous-metal	Iron and steel	Phosphorus fertilizer and phosphoric acid	Batteries	Mining	Total
1940	0.5	0.6	0.2	0.5	1	2.8
1945	0.6	0.8	0.5	0.6	1	3.5
1950	0.5	0.8	1.6	0.6	1	3.7
1955	1.4	1.3	2.0	0.7	1	6.4
1960	1.5	1.9	2.0	0.8	1	7.2
1965	2.3	2.7	2.0	0.5	1	8.5
1970	2.5	2.2	1.0	0.2	1	6.9
1975	2.5	1.4	0.5	<0.1	1	5.3
1980	1.5	0.5	0.1	<0.1	1	3.0
1985	0.8	0.1	0.1	<0.1	1	2.0
1990	0.1	0.1	<0.1	<0.1	1	1.2
Total	70	60	50	20	50	250

whole time period, which is probably an underestimation. Nevertheless, today leaching from mine waste deposits is a major source of cadmium. Approximately 60% of total mining emissions has come from one single mining area: the Falun region.

Calculated production emissions to air are shown in Table 4. The total amount in the period 1940–1990 is slightly more than 500 t. Non-ferrous metallurgical processing is totally dominant, constituting 77% of total emissions. These industrial branches are rather concentrated: emissions from non-ferrous metal production originate almost from one single plant, Rönnskär, and most secondary production is located in a few plants, e.g. Finspång and Gusum (Fig. 2).

3.2. Consumption emissions

Consumption emissions from various products containing cadmium have been calculated from the actual use for different time periods. The use of phosphorus fertilizers, which often have a significant cadmium content, has resulted in emissions to arable land. These emissions have been regarded as consumption emissions in this study.

3.2.1. Consumption emissions from various prod-

ucts containing cadmium. The diffusion of cadmium from a specific use has been calculated according to emission factors, which give the proportion of the cadmium content in a product which will be mobilized in the environment. These emissions were accounted for in the same decade as the actual use. The following factors given by Tarr and Ayres (1990) have been used:

Metallic uses (alloys):	0.001;
Batteries, pocket type:	0.02;
Batteries, sintered plate:	0.20 (cf. text);
Stabilizers:	0.15;
Plating:	0.15; and
Pigments:	0.50.

For batteries, Tarr and Ayres (1990) give a factor of 0.02 for all types. However, in Sweden the recirculation of the sintered-plate type is limited (10–30%, SOU 1990:59). Thus, in this study the same factor has been used as for mercury batteries (0.20).

A discussion of the uncertainties of these factors is presented in Bergbäck et al. (1992) and the conclusions made for lead in this paper are also relevant for cadmium.

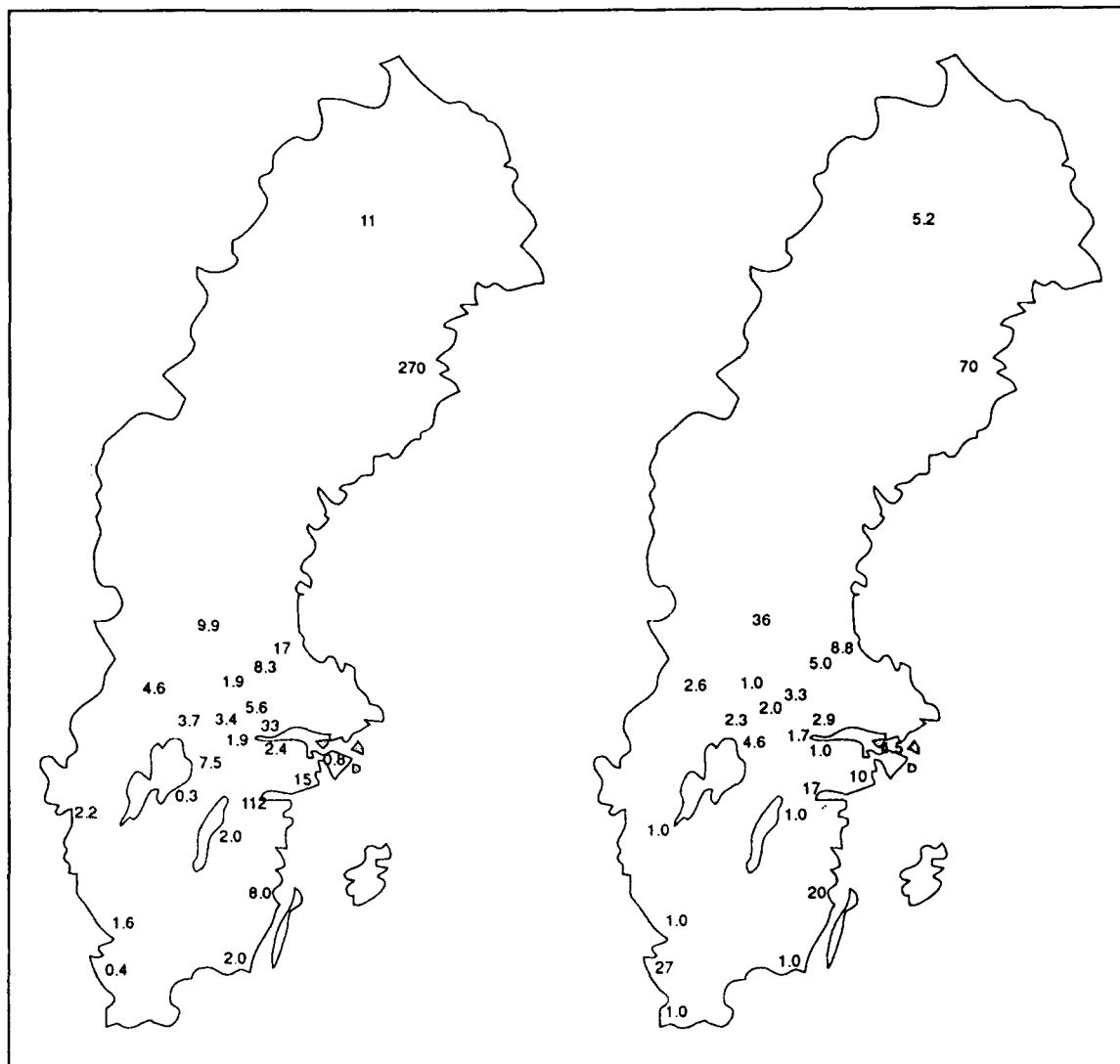


Fig. 2. Calculated total production emissions to air (left) and water (in tonnes) in Sweden from 1940 to 1990. The centre of each A-region is indicated by the estimates.

The total use of cadmium in alloys (250–1000 t) or as an impurity in zinc (20–200 t) combined with a small emission factor (0.001) has given rise to limited consumption emissions (less than 1.2 t). As these emissions should be spread over the whole area of Sweden, over 50 years, this source has not been considered. Furthermore, for pigments, the rather limited use of cadmium in artist's paint and in glass or ceramics has been excluded due to the longevity of these products.

The calculated consumption emissions are shown in Table 5 and the yearly emissions are estimated in Table 6. The total amounts to approximately 500 t over 50 years. The yearly emissions, reached a maximum in the period 1970–1980 and were still at a high level in 1990. This reflects the drastic increase in the use of Ni-Cd batteries, but also the reduced use of cadmium in pigments, stabilizers and plating since the ban on cadmium in the early 1980s. Electroplated metals have been

Table 4

Calculated cadmium emissions to air from various branches of industry in Sweden 1940–1990, based on production figures (t/year)

	Non-ferrous metal		Iron and steel	Batteries	Total
	Primary	Secondary			
1940	2.3	0.2	1.0	0.1	3.6
1945	2.9	0.3	1.1	0.1	4.4
1950	2.3	0.6	1.2	0.1	4.2
1955	7.3	0.9	1.9	0.2	10.3
1960	7.6	4.3	2.9	0.2	15.0
1965	12	8.0	4.0	0.3	24.3
1970	10	10	3.6	0.4	24.0
1975	6.0	1.0	2.8	0.1	9.9
1980	3.2	0.7	2.0	0.1	6.0
1985	2.7	0.3	1.0	<0.1	4.0
1990	0.9	0.1	0.7	<0.1	1.7
Total 1940–1990	270	140	110	8	530

the major source (36%) of the total consumption emissions, but this use is now limited to a few applications (e.g. aircraft industry). The dominant source of cadmium emissions in the 1980s was Ni-Cd batteries (66%).

3.2.2. The consumption of phosphate fertilizers containing cadmium. Estimated amounts of cadmium (in g Cd/ha) added to agricultural soils as impurities in phosphorus fertilizers has been given in Rapport från miljövårdsberedningen (1989) for the 20th century (Table 7). The development of

Table 6

Calculated consumption emissions of cadmium from various uses in Sweden 1940–1990 (t/year)

	Total consumption emissions (t/year)	
1940	3.4	
1950	4.7	average 1940–1960
1960	9.1	1950–1970
1970	15.0	1960–1980
1980	15.9	1970–1990
1990	13.8	

acreage of arable land provides the total amounts of cadmium distributed in Sweden for different decades. This total has been distributed among A-regions according to the actual acreage of arable land. The south of Sweden has been assumed to have used twice as much phosphorus fertilizers as the north (A. Andersson, Swedish University of Agricultural Sciences, pers. commun.).

3.3. Total emissions

The development of different sources of cadmium emissions is shown in Fig. 3. Production emissions, both to air and water, decreased dramatically in the 1970s, as a result of legislation and improved technology. Fertilizers have been used since the turn of this century, and taking into consideration the total amounts from 1900–1990, the relative importance of this source will obviously increase. The contribution from consumer uses

Table 5

Calculated consumption emissions of cadmium from various uses in Sweden 1940–1990

	Pigments	Stabilizers	Ni-Cd batteries		Plating	Total (t/10 years)
			Jungner	Sintered-plate		
1940–1950	—	—	1.3	—	33	34
1951–1960	7.2	4.5	2.3	—	45	59
1961–1970	37	24	3.0	—	58	122
1971–1980	63	50	4.1	17	45	179
1981–1990	13	21	3.5	88	45	138
Total	120	100	14	110	190	530

Table 7

Calculated cadmium emissions from the use of phosphorous fertilizers in Sweden 1900–1990

	g Cd/ha 5 year ^a	Cd (t/year)		Cd (t/10 years)	
1900–1905	3.0	1905	2.1		
1906–1910	3.0	1910	2.1	1901–10	21
1911–1915	3.0	1915	2.1		
1916–1920	2.0	1920	1.4	1911–20	19
1921–1925	2.5	1925	1.8		
1926–1930	4.0	1930	2.8	1921–30	20
1931–1935	4.0	1935	2.8		
1936–1940	5.0	1940	3.6	1931–40	31
1941–1945	3.5	1945	2.5		
1946–1950	7.0	1950	4.9	1941–50	37
1951–1955	10	1955	7.0		
1956–1960	10	1960	6.6	1951–60	62
1961–1965	11.5	1965	7.1		
1966–1970	14	1970	8.4	1961–70	74
1971–1975	16.5	1975	9.9		
1976–1980	7.5	1980	4.4	1971–80	76
1981–1985	5.5	1985	3.2		
1986–1990	4.5	1990	2.5	1981–90	34
Total				1900–1990	374
				1940–1990	283

^aSource: Rapport från miljöberedningen, 1989, and A. Andersson, pers. commun.

has increased drastically, mainly due to the enormous consumption of Ni-Cd batteries over the last decade. Furthermore, even taking into account the whole period studied (1940–1990), consumption

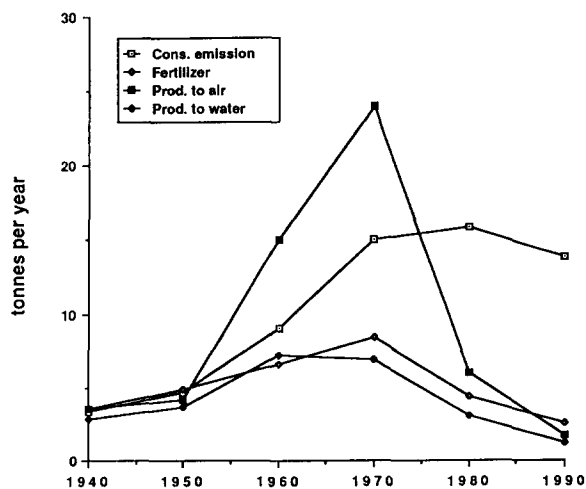


Fig. 3. Calculated cadmium emissions (t/year) from different sources in Sweden from 1940 to 1990.

emissions have been quite significant, constituting 33% of the total calculated emissions. The total amount of cadmium used in various products was approximately 3400 t for the period 1940–1990 (cadmium in alloys excluded). The emissions from these products, according to the emission factors chosen in this study, were 520 t, i.e. 15% of the total amount used. However, the remaining 85% or 2900 t have accumulated in the technosphere and constitute a future potential risk.

4. Accumulated amounts of cadmium in soil and sediments

The total calculated emissions of cadmium in Sweden for the period 1940–1990 (1900–1990 for phosphorus fertilizers) are approximately 1700 t. If this amount was to be evenly distributed over the total area of Sweden (450 000 km²) this would give an additional amount of 4 kg/km² (0.015 ppm, 0–20 cm of the top layer of soil), which is more than 10 times lower than the actual average cadmium content in Swedish soil. However, as the

use and emissions of cadmium are often restricted to specific areas, the emissions must be spatially distributed. Thus, the cadmium load has been calculated here per A-region.

4.1. Methods

Cadmium emissions, from production and consumption, will eventually end up in soil or sediment sinks. To a varying degree, cadmium absorbs to clay minerals and/or organic matter, for example, and an accumulation takes place. Here, the accumulated amounts of cadmium in soil and sediments have been calculated per decade and A-region. Total air emissions for the actual time period were distributed in relation to the areas of land and water within each A-region. Total direct water emissions, as well as air emissions deposited on the water surface, constituted the load to sediments. Consumption emissions were placed in the same category as air emissions, apart from 10% which was considered to be direct water emissions.

The transport of cadmium from soil to sediment was calculated by a simple flow model. Compared with most other heavy metals, the leachability of cadmium is relatively high. Laboratory experiments (Tyler, 1978) show that it takes 6 years (pH 4.2) for leaching to cause a 10% decrease in the total concentration of cadmium in a mor horizon. At a lower pH (3.2) the same proportion was leached in 3 years, i.e. an approximate rate of 30%/decade. Studies of metal budgets in spruce and beech forest ecosystems in southern Sweden have showed that the total supply of cadmium in the soil decreased by 0.8%/year (Bergkvist, 1986). Thus, in this region the leaching rate exceeded the total deposition. The mobility of soil cadmium is obviously closely related to the pH of the soil. The effects of experimental acidification and of liming on the transformation and the vertical distribution of cadmium in a coniferous forest soil have been studied by adding isotopically-labelled cadmium (Nömmik et al., 1984). In this study, the soil recovery of added $^{109}\text{Cd}^{2+}$ in exchangeable form decreased with the increased length of the observation period and the increased sulphuric acid application rates (Table 8). After 1 year, approximately 80% of the cadmium added was recovered with the highest leaching rate for the acidified soil. This

Table 8

Percent of added ^{109}Cd recovered in the soil profile in exchangeable form

Year after treatment	Treatment		
	Control	Acid H_2SO_4 (96 kg/ha)	Lime CaCO_3 (300 kg/ha)
1	79	75	80
2	48	30	38
3	43	32	41

Source: Nömmik et al., 1984.

tendency was accentuated in consecutive years. In addition, a redistribution of cadmium to deeper soil layers was registered. The results clearly show the high mobility of cadmium, especially in sandy podzols. However, the total amounts of cadmium leaching out from an average Swedish soil is probably less. Ongoing soil acidification, especially in southern Sweden, has probably increased leaching processes over the last few decades. However, as a first approximation, a leaching rate of 30%/decade has been used here for the whole period studied. For arable land (with a higher pH) leaching was considered not to occur, but the same proportion (30%) was assumed to follow the crops (A. Andersson, Swedish University of Agricultural Sciences, pers. commun.). Most of this proportion will eventually find its way back to arable land.

In the calculations two important simplifications were made. First, all cadmium was assumed to remain in the region in which it was emitted. Second, the contribution from foreign sources has not been taken into consideration. Instead, a state of equilibrium has been assumed with import and export counterbalancing each other. This is obviously an approximation, at least in southern Sweden where the impact from central Europe is significant. The importance of long-range transport of heavy metals has been shown by, for example, Pacyna et al. (1984). In Sweden, both deposition calculated from moss analysis and measured wet deposition exceeded the domestic emissions in the 1980s (Monitor, 1987). However, it has not been possible to reconstruct the impact and thus the net effect of foreign emission sources.

4.2 Results

Fig. 4 shows the amounts of cadmium in soil (kg/km^2) in Sweden from 1950 to 1990. Air emissions from primary (Rönnskär) and secondary (Finspång, Gусum and Skultuna) non-ferrous metal production have resulted in large amounts of cadmium in the soil. Approximately 50% of the calculated total air emissions have originated from the smelter in Rönnskär. The significance of the emission is, however, reduced by the large area

of the A-region concerned. The cadmium load in soil in these non-ferrous metal production areas culminated in 1980, i.e. the soil cadmium content decreased during the 1980s (Table 9). Obviously, this is an effect of decreased emissions and a high leaching rate. Regions with iron and steel production, but also agricultural areas in southern Sweden have had large emissions to soil, especially in the period 1950–1975. However, apart from Rönnskär and plants with secondary non-ferrous

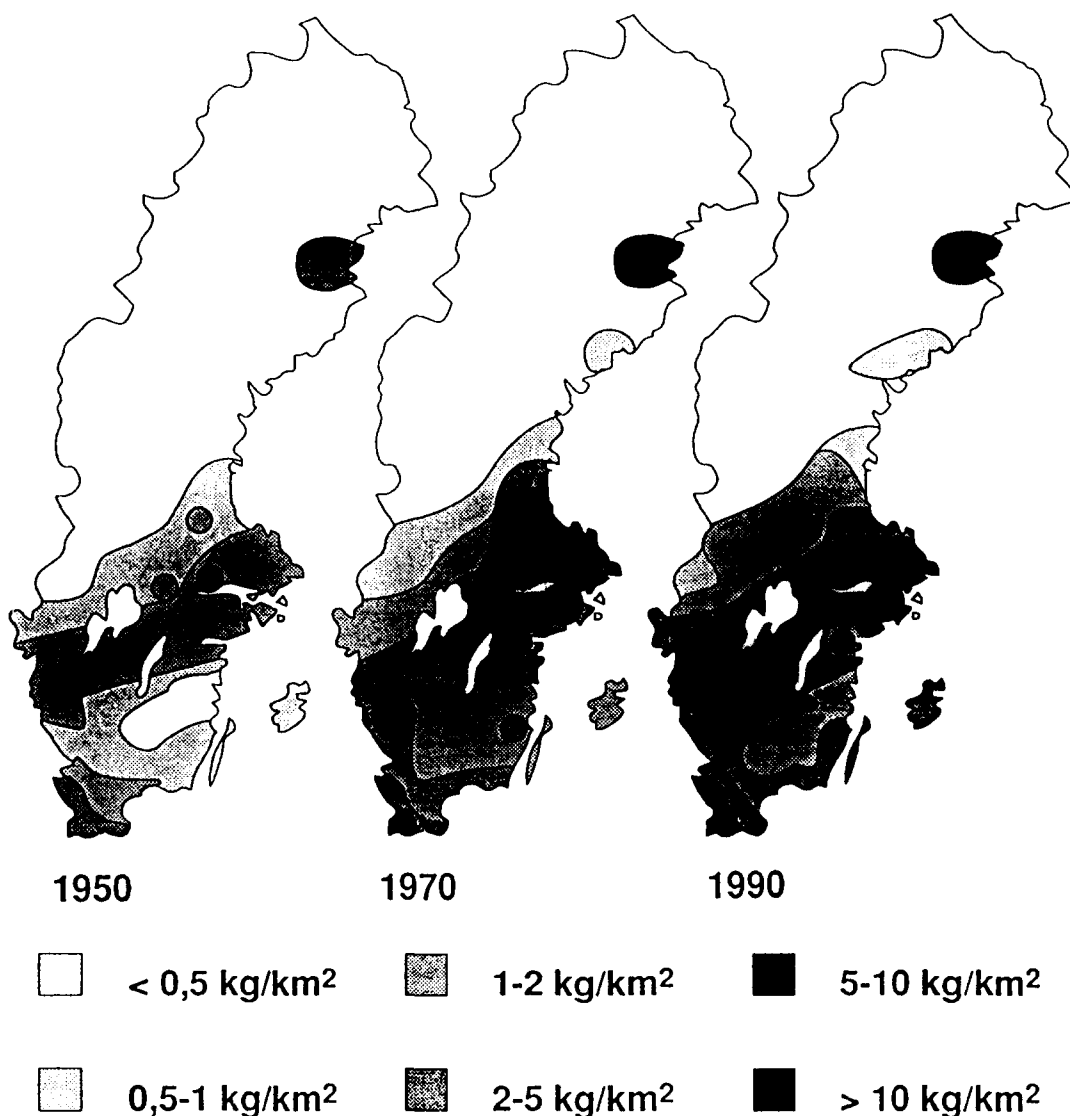


Fig. 4. Calculated anthropogenic cadmium (kg/km^2) accumulated in soils in Sweden from 1950 to 1990.

Table 9

Calculated amounts of cadmium in soil and sediments for two A-regions in Sweden (kg/km²)

	Skellefteå ^a		Norrköping ^b	
	Soil	Sediment	Soil	Sediment
1950	2.4	10.6	1.4	7.3
1960	7.0	44.3	5.3	37.5
1970	13.8	118	18.6	87.8
1980	15.4	223	21.3	165
1990	12.5	312	16.2	243

^aIncluding Rönnskär, area of A-region: 10 891 km².

^bIncluding Finsspång and Gusum, area of A-region: 4296 km².

metal production, the most dominant regions are urban areas such as Stockholm and Malmö with a high soil cadmium content caused mostly by consumption emissions. In Sweden, the cadmium content measured in relatively unpolluted soils ranges from 8 to 580 kg/km² (0.03–2.3 ppm, 0–20 cm of the top layer, soluble in HNO₃) with an average of 55 kg/km² or 0.22 ppm (Andersson, 1977). In 1990, regions with the highest calculated anthropogenic soil cadmium content had levels of 15–20 kg/km², i.e. approximately one-third of the average background level. However, deposition may cause a drastic increase in soil cadmium content when considered on a more local scale. If, for example, the total production emissions from the smelter in Rönnskär into the air (270 t) are evenly distributed within a radius of 10 km, the accumulated amounts of soil cadmium will be 600 kg/km² (2.4 ppm), assuming a leaching rate of 30%/decade. On the other hand, if 10% of the emissions are assumed to have been deposited within a radius of 1 km from the smelter industry, the soil cadmium content will be approximately 6000 kg/km² or 24 ppm. In a study from 1980, the measured cadmium concentration in the upper soil layer dropped from 22 ppm at a distance of 1.5 km to 1 ppm at 10 km (Hellström, 1980). Another example is Gusum with a total calculated cadmium emission to air of 28 t. As the calculated emission culminated in 1970–1980 (Table 9) a comparison with measurements from 1973 is of relevance (Tyler, 1975). In this study, the average

soil cadmium content at a distance of 200 m from the brass industry is approximately 40 ppm. This concentration is equivalent to a deposition of 6% of the total cadmium emitted (with the same leaching rate as before). If, as in the case of Rönnskär, 10% of the total emissions are assumed to have been deposited within a radius of 1 km, the soil cadmium content will be just above 600 kg/km² or 2.5 ppm. This corresponds well with the measured values of 2.5–4.7 ppm, at a distance of 0.5–1.2 km.

Thus, even if the calculations underestimate the impact of point sources within the A-regions, i.e. the 'local' emission landscape is not fully represented, the calculated soil cadmium content is in reasonable agreement with measured values when considered at a local scale.

Fig. 5 shows the calculated amounts of cadmium in sediment from 1950 to 1990. The emissions into water resulting from the manufacture of phosphorus fertilizers (Helsingborg, Landskrona, Norrköping), the production of Ni-Cd batteries (Fliseryd) and the production of non-ferrous metal, as well as iron and steel, have emitted large amounts of cadmium to water. The highest calculated cadmium content in sediment in 1990 (7100 kg/km²) is in the Helsingborg/Landskrona region. This is a result of large emissions from the production of phosphorus fertilizers and phosphoric acid, but also an effect of there being only a small area of water within this A-region. The water areas for some coastal A-regions are, however, underestimated in the reference used here (Atlas över rikets indelningar, 1985; Tora Friedrich, Swedish Central Bureau of Statistics, pers. commun.). Thus, the calculated sediment cadmium content in this region is probably too high. For the Malmö region (500 kg/km²) consumption emissions have been dominant, and for the Västerås region (650 kg/km²) the emissions from both non-ferrous metal production and iron and steel industry have been significant. Also, in these two regions the total area of water is small. In contrast, the largest single source of cadmium emissions to water, in Rönnskär, is a region with a large area of water. Thus, the importance of the cadmium load (300 kg/km²) in this region is somewhat underrepresented in Fig. 5.

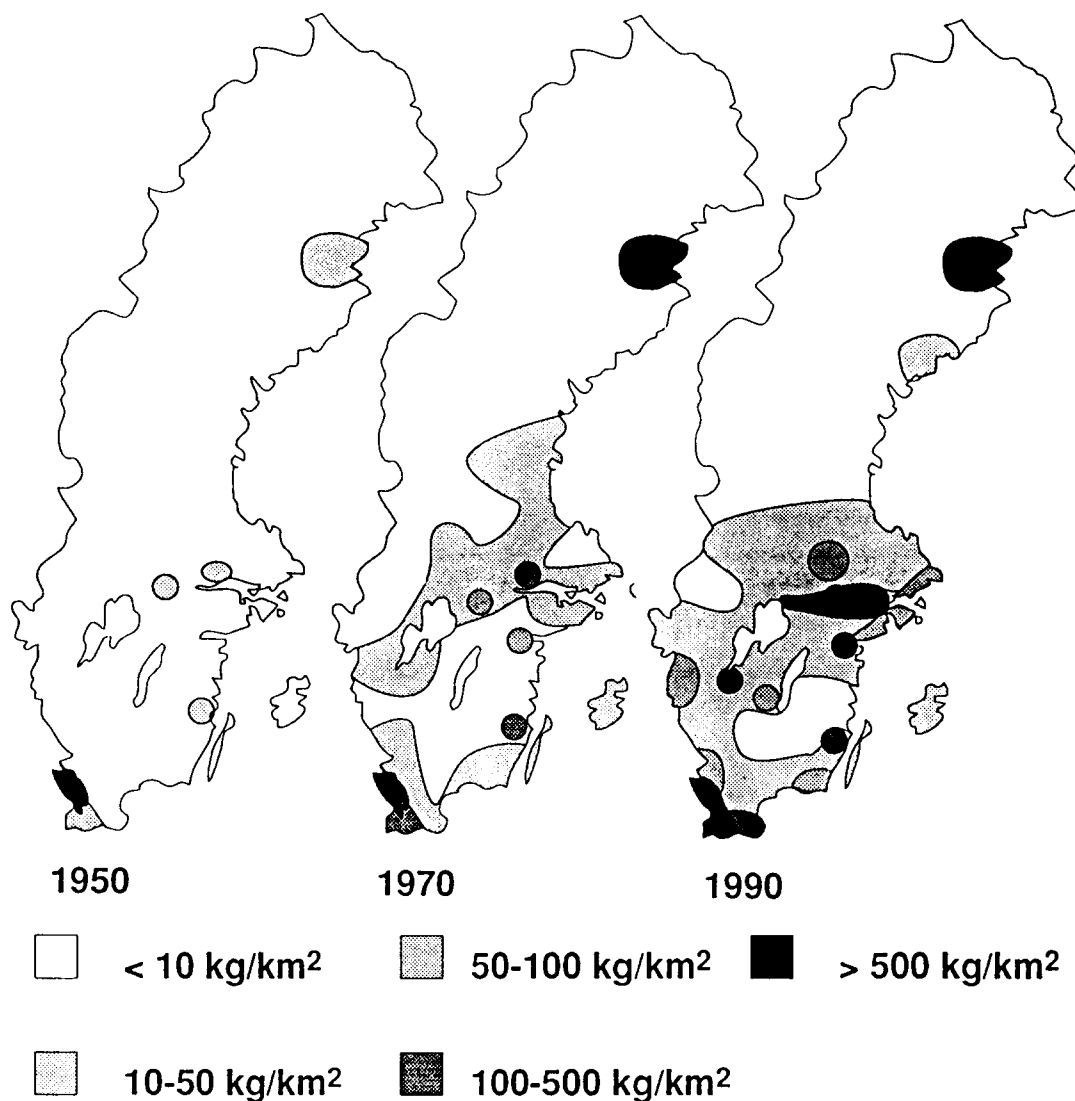


Fig. 5. Calculated anthropogenic cadmium (kg/km²) accumulated in sediments in Sweden from 1950 to 1990.

5. Conclusions

There have been dramatic changes in the relative importance of different flows of cadmium. In 1940 emissions from industry, consumption of products and the use of phosphorus fertilizers constituted approximately equal proportions of the total emissions. In 1970 industry emissions were the major source, constituting nearly 60% of the total. However, in 1990 consumption emissions were totally

dominant, constituting more than 70% of the total. This dominance was determined by the use of Ni-Cd batteries, with an estimated consumption of 90 t/year in 1990.

The rate of total emissions calculated in Sweden rose from 13 t/year in 1940 to a maximum of just over 50 t/year in 1970, and in 1990 it had fallen to 20 t/year. It is interesting to compare these anthropogenic emissions with natural 'emissions' from the weathering processes of the Swedish

bedrock. Mobilization as a result of weathering may be calculated by using the average trace metal concentrations in soil and the suspended sediment flux of 1.5×10^{16} g/year in rivers. Dissolved trace-metal flux is generally much lower than particulate flux. For cadmium, the global weathering-rate is approximately 4500 t/year (Nriagu, 1990). In Sweden's case this would mean about 13 t/year. (The area of Sweden is approximately 0.3% of the world land area.) Thus, the 'societal weathering rate' exceeded the natural rate more than 4-fold in 1970, and the present rate (1990) of anthropogenic emissions is 1.5 times higher than the natural release due to weathering.

The total calculated emissions of cadmium in Sweden, from production and consumption, have so far been approximately 1700 t. The accumulated amount of cadmium used, including cadmium in alloys and as impurities in zinc, is 5000 t. However, consumption emissions from various uses are only equivalent to 10% of the total amount used, according to the emission factors assumed in this study. The rest remain in the technosphere and, obviously, constitute a potential risk. For example, this amount of cadmium (5000 t) is equivalent to:

- the amount of cadmium in the top layer (0–5 cm) of Swedish soil;
- three times the amount of cadmium in Swedish arable land (0–20 cm); or
- a thousand times the amount of cadmium dissolved in all Swedish lakes (Nilsson and Wallgren, 1987).

Obviously, the total amount of cadmium used in Sweden is not evenly distributed, but these comparisons show clearly the magnitude of this potential pollution problem.

The use of cadmium must be restricted, especially in Ni-Cd batteries, otherwise the amount of cadmium in the technosphere will double within the next 50 years. If cadmium batteries are substituted by lithium or nickel hydride batteries, for example, the future cadmium emission rate will probably be less than the amount of cadmium released by natural weathering.

The amounts of cadmium accumulated locally

in soil and sediments are considerable. The use of phosphorus fertilizers has raised the cadmium content of arable land. This has resulted in increased cadmium concentrations during the 20th century in Swedish autumn wheat, for example (Andersson and Bingenfors, 1985). In general, the uptake of cadmium in plants and animals is related to acidification. Approximately 15% of Swedish arable land has a soil pH lower than 5.5, which facilitates the uptake of cadmium by oats, for instance (Friberg et al., 1990). In acidic soils, the mobilisation of heavy metals, and especially cadmium, is increased. Eventually this will result in raised cadmium concentrations in the groundwater, causing problems for more than 0.5 million Swedes who get their drinking water from wells in areas with acidic soils (Jacks and Kucera, 1982).

In the simple flow model used in this study, cadmium is assumed to be accumulated in soil or sediment sinks. However, the transport of cadmium from soil to sediments is often governed by slow movements of the groundwater. Thus, the calculated amounts of accumulated cadmium in sediments are probably overestimated and a significant proportion of the cadmium can still be found in the groundwater.

Man's present exposure to cadmium is close to levels that are detrimental to health (Friberg et al., 1986). Thus, in areas with a low soil buffer-capacity, cadmium may become a major future pollution problem. In addition, the change in land-use as a result of abandoning arable land and discontinuing liming, will constitute a major cause of soil acidification. Agricultural soils have often been the recipient of cumulative doses of heavy metals over long periods, and a sudden decline in pH could trigger the release of cadmium.

Finally, even if the future societal weathering rate is less than the natural rate, cadmium will continue to be a health problem until the accumulated amounts of soil cadmium have eventually been immobilised in sediments.

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